Estimation Theory: Coursework Submission 2

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Contents

1 Introduction **1**
1.1 Motivation 2
1.2 Contributions 2
1.3 Paper Organisation 2

2 Background **2**
2.1 Numerical Weather Prediction 2
2.2 Deep Learning Approaches 2
2.3 Transformer Models 2

3 Method **3**
3.1 System Model 3
3.2 Beacon Placement 4

4 Results **4**

5 Discussion **4**
5.1 Limitations 4
5.2 Future work 4

Abstract

Undecided on if this is necessary.

1 Introduction

Climate prediction remains one of the most challenging problems in modern science **?**. Traditional numerical models struggle to capture the full complexity of atmospheric dynamics.

*Repository: github.com/Gijinkstra/estimation-theory-car-tracker

1.1 Motivation

Recent advances in deep learning offer a promising alternative to classical approaches **?**. This work investigates whether transformer-based architectures can outperform existing baselines on standard benchmarks.

1.2 Contributions

The main contributions of this paper are:

- A novel transformer architecture adapted for climate data.
- A benchmark comparison across five standard datasets.
- Open-source code and pretrained model weights.

1.3 Paper Organisation

The remainder of this paper is structured as follows. Section 2 reviews related work. Section **??** describes the proposed method. Section **??** presents experimental results. Section **??** concludes.

2 Background

2.1 Numerical Weather Prediction

Numerical weather prediction (NWP) has been the dominant paradigm since the 1950s These models solve partial differential equations governing atmospheric dynamics on discrete spatial grids.

2.2 Deep Learning Approaches

Several deep learning methods have been proposed for weather forecasting. can match NWP skill scores at short lead times. Subsequently, using a cubed-sphere projection.

2.3 Transformer Models

The transformer architecture has achieved state-of-the-art results across a wide range of domains. Its self-attention mechanism allows modelling of long-range dependencies, making it well suited to spatiotemporal data.

3 Method

3.1 System Model

The system model utilised for the trajectory is based on the Frenet frame of reference [INSERT REFERENCE HERE]:

$$\begin{aligned} s_{t+1} &= s_t + h\, v_{s,t} \\ l_{t+1} &= l_t + h\, v_{l,t} \\ v_{s,t+1} &= v_{s,t} + w_{s,t} \\ v_{l,t+1} &= v_{l,t} + w_{l,t} \end{aligned} \tag{1}$$

where s_t is the arc position, l is the lateral deviation from the centre line of the track, v_s is the arc velocity, v_l is the lateral velocity, and w_t is a gaussian noise term:

$$w_t \sim \mathcal{N}(0, \begin{bmatrix} 0.2 \\ 0.01 \end{bmatrix}) \tag{2}$$

,effectively acting as small deviations in acceleration. The state vector is:

$$\mathbf{x_t} = \begin{bmatrix} s \\ l \\ v_s \\ v_l \end{bmatrix} \tag{3}$$

The full matrix form implemented in MATLAB is as follows:

$$\mathbf{x}_{t+1} = \mathrm{A}\mathbf{x}_t + \mathrm{G}\mathbf{w}_t \tag{4}$$

where $\mathbf{x}_t$ is the state vector, A is the state transition matrix, G is the noise transition matrix. Note that the Gaussian noise term acts independantlyon the velocity components of the system. No uncertainty in position has been utilised in this model.

This state model is utilised to keep the dynamics of the system linear. By changing the co-ordinate basis of the state model, this shifts all the non-linearity into the measurement model of the system. As the problem is inherently non-linear, the Jacobian that was created in earlier steps of the development process can be modified to include the conversion from the current position in the Frenet frame back to Cartesian co-ordinates for calculation.

$$\mathbf{y}_t = ||r_{x,y} - r_{x,y}^{(i)}||_2 + o_t \tag{5}$$

where $\mathbf{y}$ is the output vector (Euclidean distance from each beacon), $r_{x,y}$ is the position of the vehicle on the track in cartesian co-ordinates, $r_{x,y}^{(i)}$ is the position of each beacon measurement in cartesian co-ordinates and o_t is a gaussian noise term. Noise is added independantly to each beacon reading, with a standard deviation of 1.5m as defined in the specifcation.

There are defined physical limitations on the parameters of the vehicle as follows:

$$\begin{aligned} l &\in [-2, 2] \\ v_s &\in (0, 25) \\ v_l &\in (-0.55, 0.55) \end{aligned}$$

3.2 Beacon Placement

Location must be inferred from a minimum number of beacon measurements as outlined by the specification. A single beacon measurement is not sufficient to infer the position of the vehicle, as the Euclidean distance would only provide a radial distance from the singular beacon (the distance from the beacon could be anywhere on a circle surrounding the beacon).

Two beacons would present a localisation problem, as the location of the vehicle could be at either of two intersection points on the radial distances from the beacon positions.

Three beacons are the minimum number required to completely localise the position of the vehicle on the track geometry. Given this, the positioning of the beacons are considered to ensure that there is no bias to any one location on the track. As the beacon readings are available immediately and there is no increase in the additive noise variance to the system based on the range of the beacons (the further away the beacons, the greater the noise), beacons can be placed further away to minimise the relative noise contribution.

4 Results

5 Discussion

Blah

5.1 Limitations

Blah blah

5.2 Future work

Did you see what I wrote in Sec. **??**? Not too controversial I hope!

Figure 1: Mr Rizzler.